Adagio Institute, Inc.
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First-Round Questions to the Candidate

Paper 3

Convex Perturbation and the Suppression of the Indivisibility Deficit

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Degree:	Doctor of Philosophy in Mathematical Physics (Concentration in Stochastic Quantum Theory), by Published Works
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Independent evaluation:	Dr S. R. Ismail-Sutton, Durham University
Identifiers:	Q-P03-01 to Q-P03-08 (technical report); Q-PF-08, Q-PF-09 (portfolio-level evaluation)
Compiled:	14 October 2026

This document belongs to the examination record (03_examiner_questions). It lists the questions put to the candidate in the first round of the examination of Paper 3 and gives each a stable identifier, by which the Report, the Technical Annex and the correspondence can cite it.

1 Sources and identifiers

Technical report. Section 2 transcribes Section 7, "Questions to the author", of the Doctoral Examiner's technical report on Paper 3, `TR-P03-DOCv1-EX01-Rv1.pdf`. The report identifies the version examined as SSRN, DOI 10.2139/ssrn.6812461, 17 July 2026, and carries the evaluation date 4 September 2026. The Doctoral Examiner sent it to the candidate on 8 September 2026 (COR-P03-01), and the candidate answered, to both examiners, on 13 September 2026 (COR-P03-03).

Identifiers. `Q-P03-`k is Question k of that report. The candidate's letters and the Technical Annex cite the questions by these numbers. The text is the report's, word for word; its section signs (§) refer to sections of the paper, and its finding numbers to findings of the report. Notes in small type are not part of the questions. They identify unresolved references, later corrections and related questions.

Independent evaluation. The independent evaluator's review of Paper 3 (`TR-P03-DOCv1-EX02-Rv1.pdf`, Exhibit J, 02_technical_evaluations) states its points as comments and poses no questions. It was sent to the candidate on 4 September 2026 and again on 11 September, and is filed as COR-P03-02; the candidate answered it together with the technical report on 13 September 2026 (COR-P03-03). It is not transcribed here.

Portfolio-level evaluation. Section 3 transcribes the questions of the portfolio-level technical evaluation (version 2, 4 September 2026; Exhibit H, Section 7, 02_technical_evaluations) that concern Paper 3. `Q-PF-`k is Question k of that evaluation and has the same identifier in all six documents; Q-PF-09 concerns all six papers and appears in each. The evaluation names the works W1–W6, and Wn is Paper n.

2 Questions of the technical report

Source: `TR-P03-DOCv1-EX01-Rv1.pdf`, Section 7, Questions 1–8.

Q-P03-01 Give a stochastic-process realization producing the same constant g in every two-time conditional kernel, or explain why none is required and recast (1) as a formal perturbation path. State explicitly whether the families are assumed process-induced, and what remains valid if they are abstract two-time stochastic families instead.

Note. (1) is the display of the report, Section 1: $\Gamma_g(t \leftarrow s) = (1-g)\,\Gamma_0(t \leftarrow s) + g\,\Gamma_{\mathrm{ref}}(t \leftarrow s)$, $g \in [0,1]$, the convex perturbation that the paper studies.

Q-P03-02 Explain the mathematical distinction between your terminal threshold and the robustness measures of Anand & Brun and of De Santis & Giovannetti, and add both references. The distinctions I consider defensible are set out in Finding 5; confirm or replace them.

Q-P03-03 Correct the assertion that exact restoration occurs "that is, at $g = 1$". Can Γ_g become exactly divisible at an interior coupling? Your own §4.3 example says it can; state what, if anything, characterises the set of such couplings.

Q-P03-04 Reconcile the disclaimer that "observer coupling" carries no interpretive weight with the interpretive conclusions drawn in §7.5.

Q-P03-05 Qualify the no-generator statement: which generator classes are actually excluded by CK failure?

Q-P03-06 How would ε be calibrated from finite-sample transition estimates, and what null hypothesis would be tested?

Q-P03-07 What invariant meaning, if any, should be attached to the reference-dependent interaction scale M_*? Can the terminal bound be sharpened using operator direction rather than a sign-blind norm bound? Remark 4.5 reports a factor-of-eight gap in your own example, which suggests there is room.

Q-P03-08 Keep the speed-limit comparison strictly analogical, as §7.3 already does, and state whether you intend to pursue the rate-form question or to withdraw it.

3 Questions of the portfolio-level evaluation

Source: portfolio-level technical evaluation, version 2, 4 September 2026 (Exhibit H, 02_technical_evaluations), Section 7. Question 8 concerns Paper 3; Question 9 concerns all six papers.

Q-PF-08 **W3.** Reconcile the disclaimer that "observer coupling" carries no interpretive weight with the interpretive conclusions of §7.5. Separately: what dynamical mechanism yields a convex mixture of transition kernels at fixed weight?

Note. Q-P03-04 asks the first part; Q-P03-01 bears on the second.

Q-PF-09 **Deposit venue.** State whether the works will be deposited on arXiv (math-ph / math.PR / quant-ph).